

Experiment 7: Capture Video from Webcam and Display in Different Speeds

Aim

To capture video from a webcam using Python and display the video in normal, slow motion, and fast motion.

Algorithm

1. Import the OpenCV library.
2. Capture live video from the webcam using `cv2.VideoCapture(0)`.
3. Set the playback delay for slow, normal, and fast modes.
4. Read and display each video frame continuously.
5. Press 'q' to stop the video capture.
6. Release the webcam and close all display windows.

Python Code

```
import cv2

mode = "normal"    # Change to "slow" or "fast"

delay = {
    "slow": 100,
    "normal": 30,
    "fast": 5
}

cap = cv2.VideoCapture(0)

while True:
    ret, frame = cap.read()

    if not ret:
        break

    cv2.imshow("Webcam Video", frame)

    if cv2.waitKey(delay[mode]) & 0xFF == ord('q'):
        break
```

```
cap.release()  
cv2.destroyAllWindows()
```

Result

The live video was successfully captured from the webcam and displayed in normal, slow motion, and fast motion using Python.